Cheng Guo

CONTACT INFORMATION

Website: https://chengg04.github.io *E-mail:* cguo2@clemson.edu

RESEARCH INTERESTS

- Methodologies: Mixed-integer linear and nonlinear programming, Conic programming, Stochastic and robust optimization, Game theory, Decomposition algorithms
- Application areas: Power systems, Market design and pricing, Healthcare.

EXPERIENCE

Clemson University, Clemson, SC December 2021 - present
School of Mathematical and Statistical Sciences
Assistant Professor, area: Operations Research (tenure clock start: Fall 2022)

Columbia University, New York, NY 2021 - 2022
Department of Industrial Engineering and Operations Research
Visiting Researcher
- Host: Daniel Bienstock
- DOE ARPA-E PERFORM (Performance-based Energy Resource Feedback, Optimization, and Risk Management) project

EDUCATION

University of Toronto, Toronto, ON September 2017 - November 2021
Department of Mechanical and Industrial Engineering
Ph.D. in Industrial Engineering, GPA: 3.96/4.00
- Advisor: Merve Bodur
- Selected courseworks: Stochastic Programming & Robust Optimization, Modeling Interactions on Networks, OM Matching Markets, Mathematical Methods in Power Systems

Columbia University, New York, NY 2015 - 2017
Department of Industrial Engineering and Operations Research
M.S. in Operations Research
- Selected courseworks: Transportation Analytics & Logistics, Optimization I, Programming for Financial Engineering, Seminar on Queueing Theory

Wuhan University, Wuhan, China 2011 - 2015
School of Economics and Management
B.A. in Economics, B.S. in Mathematics
- Hongyi Honor Program Outstanding Graduates Award
- Selected courseworks: Advanced Microeconomics, Industrial Organization, Advanced Financial Theory, Advanced Econometrics, Chaotic Dynamical Systems, Topology

PUBLICATIONS

(*: corresponding author; Underline: student coauthor)

6. *Networked Markets with Production and Edge Capacity Constraints: From Competitive Equilibria To Market Entry* [full version]

Cheng Guo*, Jiayi Wang, Ozan Candogan
ACM Conference on Economics and Computation (EC) (2026)

5. *Copositive Duality for Discrete Energy Markets* [pdf]
Cheng Guo*, Merve Bodur, Joshua A. Taylor
Management Science 72.3 (2026): 2022-2040

4. *Tightening Quadratic Convex Relaxations for the Alternating Current Optimal Transmission Switching Problem* [pdf]
Cheng Guo*, Harsha Nagarajan, Merve Bodur
INFORMS Journal on Computing, 38.3 (2026): 921-942
• **Appeared as a featured article in the May-June 2026 issue.**

3. *Risk-Aware Security-Constrained Unit Commitment: Taming the Curse of Real-Time Volatility and Consumer Exposure* [pdf]
Daniel Bienstock, Yury Dvorkin, Cheng Guo*, Robert Mieth, Jiayi Wang
IEEE Transactions on Energy Markets, Policy and Regulation 2.4 (2024): 536-551

2. *Generation Expansion Planning with Revenue Adequacy Constraints* [pdf]
Cheng Guo*, Merve Bodur, Dimitri J. Papageorgiou
Computers & Operations Research 142 (2022): 105736

1. *Logic-based Benders Decomposition and Binary Decision Diagram Based Approaches for Stochastic Distributed Operating Room Scheduling* [pdf]
Cheng Guo*, Merve Bodur, Dionne M. Aleman, and David R. Urbach
INFORMS Journal on Computing 33.4 (2021): 1551-1569

Preprints

3. *Networked Cournot Competition under Congestion* [pdf]
Cheng Guo*, Jiayi Wang, Ozan Candogan, submitted
• Conference version accepted at EC 2026

2. *Pricing Discrete and Nonlinear Markets With Semidefinite Relaxations* [pdf]
Cheng Guo, Lauren Henderson, Ryan Cory-Wright, Boshi Yang

1. *Incentivizing Investment and Reliability: A Study on Electricity Capacity Markets* [pdf]
Cheng Guo, Christian Kroer, Yury Dvorkin, Daniel Bienstock

Honors and Awards

- Finalist for student Anna Deza, INFORMS Undergraduate OR Prize Competition, 2020
- Mixed Integer Programming Workshop Student Travel Support, 2019
- Hongyi Outstanding Graduates Award, 2015
- Economics and Management School Scholarship, 2013 - 2014

Funding

- Clemson University - Clemson Faculty SUCCEEDS: Program 1 (Project Initiation/SEED Funding, awarded to 16 PIs university-wide) (PI), 2024
- University of Toronto - Bert Wasmund Graduate Fellowships in Sustainable Energy Research, 2018

Teaching

Clemson University
Instructor

- MATH 8100 - Mathematical Programming (graduate): Fall 2022, Spring 2023, Spring 2024, Fall 2024, Fall 2026
- MATH 4400/6400 - Linear Programming (undergraduate/graduate): Fall 2024, Fall 2025, Spring 2026
- STAT 3090 - Introductory Business Statistics (undergraduate): Spring 2022 (virtual), Fall 2023, Fall 2024, Fall 2025, Fall 2026

University of Toronto
Tutorial Teaching Assistant

- MIE 562 - Scheduling (undergraduate/graduate): Fall 2019, Fall 2020
- MIE 335 - Algorithms and Numerical Methods (undergraduate): Winter 2019

Wuhan University
Teaching Assistant

- Probability Theory (undergraduate): Fall 2014

Advising

Ph.D. Students
Lauren Henderson, Benjamin Hamlin (2026, co-advised with Margaret Wiecek)

M.S. Students
Lauren Henderson (2024)

Undergraduate Students
Jiayi Wang (co-advised, Columbia B.S. 2022 → Stanford Ph.D.), Anna Deza (co-advised, U. Toronto B.A.Sc. 2020 → UC Berkeley Ph.D.), Ryan Do (co-advised, U. Toronto B.A.Sc. 2019 → U. Toronto M.Eng.)

Ph.D. Thesis Committee Member
Kristen Joyce, Sarah Kelly (2024)

M.S. Thesis Committee Member
Yunheng Jiang (2022)

Invited Talks

- Ohio State University, ISE Seminar, Columbus, OH — October, 2026
- MIT, EIMC2 Seminar, Boston, MA — July, 2026
- Cornell University, FIND Seminar, Ithaca, NY — April, 2024
- Mixed Integer Programming Workshop, Los Angeles, CA — May, 2023
- Polytechnique Montreal, GERAD Seminar, Virtual — May, 2022
- Discrete Optimization Talks, Virtual — December, 2020

Conference Presentations

- INFORMS Annual Meeting, San Francisco, CA — November, 2026
- INFORMS MSOM Conference, Boston, MA — July, 2026
- Marketplace Innovation Workshop (MIW), Online — May, 2026
- Production and Operations Management Society (POMS) Conference, Reno, NV — May, 2026

- INFORMS Optimization Society Conference, Atlanta, GA — March, 2026
- INFORMS Annual Meeting, Atlanta, GA — October, 2025
- IEEE Power and Energy Society (PES) General Meeting, Austin, TX — July, 2025
- International Conference on Continuous Optimization (ICCOPT), Los Angeles, CA — July, 2025
- Production and Operations Management Society (POMS) Conference, Atlanta, GA — May, 2025
- INFORMS Computing Society Conference, Toronto, ON — March, 2025
- INFORMS Annual Meeting, Seattle, WA — October, 2024
- International Symposium on Mathematical Programming, Montreal, QC — July, 2024
- INFORMS Optimization Society Conference, Houston, TX — March, 2024
- INFORMS Annual Meeting, Phoenix, AZ — October, 2023
- INFORMS MSOM Conference, Montreal, QC — June, 2023
- INFORMS Annual Meeting, Indianapolis, IN — October 2022
- International Conference on Continuous Optimization (ICCOPT), Bethlehem, PA — July, 2022
- INFORMS Optimization Society Conference, Greenville, SC — March, 2022
- INFORMS Annual Meeting, Virtual — October 2021
- International Conference on Game Theory (poster), Virtual — July 2021
- IPCO Conference (poster), Virtual — June 2021
- CORS Annual Conference, Virtual — June 2021
- Mixed Integer Programming Workshop (poster), Virtual — May 2021
- Grid Science Winter School (poster), Virtual — January 2021
- INFORMS Annual Meeting, Virtual — November 2020
- INFORMS Annual Meeting, Seattle, WA — October 2019
- DIMACS Workshop on MINLP (poster), Montreal, QC — October 2019
- Mixed Integer Programming Workshop (poster), Boston, MA — July 2019
- Optimization Days, Montreal, QC — May 2019
- INFORMS Computing Society Conference, Knoxville, TN — January 2019

Academic Service

- Journal reviewer for *Operations Research, Mathematical Programming, Management Science, Manufacturing & Service Operations Management, SIAM Journal on Optimization, INFORMS Journal on Computing, INFORMS Journal on Data Science, Transportation Science, Production and Operations Management, Computers and Operations Research, IEEE Transactions on Power Systems, IEEE Transactions on Energy Markets, Policy and Regulation*
- Conference reviewer for *ACM Conference on Economics and Computation (EC) (2026)*, *IEEE Power & Energy Society General Meeting (PES) (2024 - 2026)*
- Award committee member for INFORMS ENRE Best Publication Award - Energy (2026), MIP Workshop poster competition (2026)
- Program committee member for Mixed Integer Programming (MIP) Workshop (2026)
- Co-host for the Energy, Market, and Optimization (EMO) Seminar [website]
- Vice President/President-Elect (2025-2026) and President (2026-2027) for INFORMS Junior Fac-

ulty Interest Group (JFIG)

- Session chair for INFORMS Optimization Society Conference (2022, 2024, 2026); INFORMS Annual Meeting (2019, 2021, 2022, 2023, 2024, 2026); CORS Annual Meeting (2021)

University Service

- Clemson University SMSS Research Committee (2023-present)
- Co-organizer of University of Toronto MIE UTORG Seminar (2019-2020)

Advanced Training

- "Cultivating an Inclusive Classroom Environment", Academic Impressions 2024

Skills

- Programming language: Python, Julia, C++
- Software: Gurobi, CPLEX, Knitro, Mosek

Other Activities

- INFORMS UofT Student Chapter (Honorable Mention, 2020), Vice President (2019-2021)
- Columbia IEOR Mentorship Program, Mentor (2018-2020)
- Wuhan U. Women Soccer Team, Captain (2013-2015)